

## Ex No: 7:

### Table Creation:

```
CREATE TABLE Authors (AuthorID INT PRIMARY KEY AUTO_INCREMENT,  
VARCHAR(50), LastName VARCHAR(50));
```

```
CREATE TABLE Books (BookID INT PRIMARY KEY AUTO_INCREMENT, Title  
VARCHAR(100), Genre VARCHAR(50), Publication Year INT);
```

```
CREATE TABLE BookAuthors (BookID INT, AuthorID INT, PRIMARY KEY  
(BookID, AuthorID), FOREIGN KEY (BookID) REFERENCES Books(BookID),  
FOREIGN KEY (AuthorID) REFERENCES Authors (AuthorID));
```

```
CREATE TABLE Borrowers (BorrowerID INT PRIMARY KEY AUTO_INCREMENT,  
FirstName VARCHAR(50), LastName VARCHAR(50), MembershipDate DATE);
```

```
CREATE TABLE BorrowedBooks (BorrowerID INT, BookID INT, BorrowedDate  
DATE, DATE, PRIMARY KEY (BorrowerID, BookID), FOREIGN KEY (BorrowerID)  
REFERENCES Borrowers (BorrowerID), FOREIGN KEY (BookID) REFERENCES  
Books(BookID));
```

### Authors

| AuthorID | FirstName | LastName |
|----------|-----------|----------|
| 1        | George    | Orwell   |
| 2        | J.K.      | Rowling  |
| 3        | Mark      | Twain    |

### Books

| BookID | Title                                    | Genre     | PublicationYear |
|--------|------------------------------------------|-----------|-----------------|
| 1      | 1984                                     | Dystopian | 1949            |
| 2      | Harry Potter and<br>the Sorcerer's Stone | Fantasy   | 1997            |
| 3      | The Adventures of<br>Tom Sawyer          | Fiction   | 1876            |

### BookAuthors

| BookID | AuthorID |
|--------|----------|
| 1      | 1        |
| 2      | 2        |
| 3      | 3        |

### Borrowers

| BorrowerID | FirstName | LastName | MembershipDate |
|------------|-----------|----------|----------------|
| 1          | Alice     | Johnson  | 2022-03-15     |
| 2          | Bob       | Smith    | 2023-07-01     |
| 3          | Clara     | Davis    | 2021-11-20     |

### BorrowedBooks

| BorrowerID | BookID | BorrowedDate | ReturnDate |
|------------|--------|--------------|------------|
| 1          | 2      | 2024-01-10   | 2024-01-25 |
| 2          | 1      | 2024-02-05   | 2024-02-20 |
| 3          | 3      | 2024-03-01   | 2024-03-15 |